

Md Zesun Ahmed Mia

Specialist in ML for Yield Optimization & Process Control

Focused on leveraging data and AI to improve semiconductor fabrication processes. Combines a deep understanding of ML with practical knowledge of manufacturing workflows.

Education

- 08/2022–05/2027 **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00*
(Expected) Advisor: Dr. Abhronil Sengupta
- 08/2022–08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00* | Thesis: Neuromorphic Computing for Lifelong Learning
- 02/2015–04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, *CGPA: 3.81/4.00*

Industrial Experience

- 05/2025–07/2025 **Graduate Technical Intern**, *Intel Corporation*, Hillsboro, OR
- Developed AI-driven predictive models for process yield optimization and critical dimension (CD) control in semiconductor manufacturing, enabling automated recipe optimization.
 - Implemented machine learning algorithms to predict optimal deposition parameters and automate recipe generation for high-volume semiconductor production.
 - Designed comparative analysis framework between AI-based models and statistical models (JMP) for enhanced prediction accuracy and process window optimization.
 - Conducted comprehensive data analysis using process characterization data to validate model performance and establish robust process control limits.
- 09/2020–07/2021 **R&D Engineer**, *SEMWAVES Ltd.*, London, UK, part-time
- Delivered a 50 kW hybrid renewable system (solar + smallscale hydro) for an offgrid Bangladeshi site, supporting reliable community power.
 - Owned technical leadership and coordination (vendors, field teams, stakeholders); directed device selection, integration, QA/safety procedures, and optimization against load profiles and uptime targets.

Technical Skills

ML for Semi-conductors	Yield prediction, defect detection, process monitoring, EDA automation, ML/EDA integration, process-aware modeling, reliability analysis, AI accelerator development, HPC applications, parallel computing architectures
ML Tools	ONNX, TensorRT, PyTorch, TensorFlow, scikit-learn, Keras, Data Visualization (Matplotlib, Seaborn, Plotly), Pandas, NumPy, JMP, Jupyter, LaTeX
Hardware, EDA & Sim.	PCB design, Cadence Virtuoso, Spectre, HSPICE, TCAD, COMSOL, ModelSim, Synopsys (Design Compiler, PrimeTime, VCS), Oscilloscope, Signal Generator, CIM (Compute-In-Memory) Systems, Systolic Arrays, PIM, near-memory computing, memory-centric computing, data flow optimization
Programming	Python, CUDA, MATLAB, Verilog, Shell, C/C++, Bash, Linux/Unix
DTCO & STCO	Design-technology and system-technology co-optimization, cross-layer PPA, process-aware design, system-architecture codesign
Fabrication	Lithography, Etching, Deposition (CVD, HDP-CVD, PVD), E-beam, SEM, AFM, DSIMS, FTIR
AI Dev.	Generative and agentic AI tools (Cursor, Copilot, etc.) for research, design, and code development
Collaboration	Git, Microsoft Office, Google Workspace

Academic Research and Teaching Experience

- 08/2022– **Graduate Research Assistant**, *Penn State*, University Park, PA
- Present
- Developed compact models of FeFETs in Verilog-A and MATLAB from characterization data, enabling their integration into standard EDA toolchains for circuit design and verification.
 - Designed an ML-based workflow to assess the system-level impact of fabricated spintronic device variations, enabling faster feedback for process optimization and EDA automation.
- 08/2024– **Graduate Teaching Assistant**, *Penn State*, State College, PA
- 05/2025
- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
 - Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.
- 02/2021– **Lecturer**, *University of Liberal Arts Bangladesh (ULAB)*, Dhaka, Bangladesh
- 08/2022
- Taught undergraduate courses: Solid State Devices, Digital Circuit Design, Semiconductor Device Physics, Power Electronics.
 - Developed lab modules and supervised projects on semiconductor devices and circuits.

Projects

- Astromorphic Transformer** Lead Student Researcher, 2022–2025. Developed a bioplausible transformer architecture leveraging neuron-astrocyte interactions to emulate self-attention mechanisms. Incorporated Hebbian and presynaptic plasticities with non-linearities and feedback, achieving superior accuracy and faster convergence on sentiment classification (IMDB), image classification (CIFAR-10), and language modeling (WikiText-2) tasks. See publication: [IEEE TCDS 2025].
- Spintronic Device-Based Memory** Lead Student Researcher, 2023–Present. Led end-to-end fabrication of spintronic memory arrays using e-beam lithography, sputtering deposition, and lift-off patterning. Performed comprehensive electrical characterization (I-V, R-H loops, retention, endurance) and extracted device parameters for compact model development. Advanced non-volatile memory technologies for neuromorphic computing and in-memory processing applications.
- Neuromorphic Cybersecurity with Lifelong Learning** Lead Student Researcher, 2023–2025. Developed a Hierarchical Dynamic Spiking Neural Network (D-SNN) for Network Intrusion Detection Systems combining static SNN detection (94.3% accuracy) with adaptive dynamic SNN classification using GWR-inspired structural plasticity and novel Adaptive STDP (Ad-STDP) learning. Achieved 85.3% overall accuracy on UNSW-NB15 benchmark in semi-supervised lifelong learning, demonstrating superior adaptation to new attack types while mitigating catastrophic forgetting (5.3% improvement over baseline). See publication: [arXiv], [ICONS 2025].

Select Publications

- [1] **Md Zesun Ahmed Mia**, Malyaban Bal, Sen Lu, George M. Nishibuchi, Suhas Chelian, Srinivasan, and Abhronil Sengupta. “Neuromorphic Cybersecurity with Semi-Supervised Lifelong Learning”. In: *Proceedings of the International Conference on Neuromorphic Systems (ICONS '25)*. IEEE Press, 2025, pp. 235–238. DOI: 10.1109/ICONS69015.2025.00043.
- [2] **Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. “Delving deeper into astromorphic transformers”. In: *IEEE Transactions on Cognitive and Developmental Systems* (2025).
- [3] KM Ashraful Hoque Fahim, Md Jubair Hasan Khalid, **Md Zesun Ahmed Mia**, and Mirza Rasheduzzaman. “Study of 3-nm Cylindrical GAAFETs with Variations in High-k Dielectric Gate-oxide Materials”. In: *2022 IEEE Symposium on Industrial Electronics & Applications (ISIEA)*. IEEE. 2022, pp. 1–5.
- [4] Md Moinul Islam, Monjurul Haque, Saiful Islam, **Md Zesun Ahmed Mia**, and SMA Mohaiminur Rahman. “DCNN-LSTM based audio classification combining multiple feature engineering and data augmentation techniques”. In: *Intelligent Computing & Optimization: Proceedings of the 4th International Conference on Intelligent Computing and Optimization 2021 (ICO2021) 3*. Springer. 2022, pp. 227–236.

Recognitions

- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024), Milton and Albertha Langdon Memorial Fellowship (2023), Melvin P. Bloom Memorial Fellowship (2022)

Professional Affiliations

- Reviewer, IEEE TNNLS (2025), Design Automation Conference (DAC) (2025), IEEE MWSCAS (2025)
- Student Member, IEEE (2015–Present), Executive Member, EDS, IEEE BD (2021–2022)